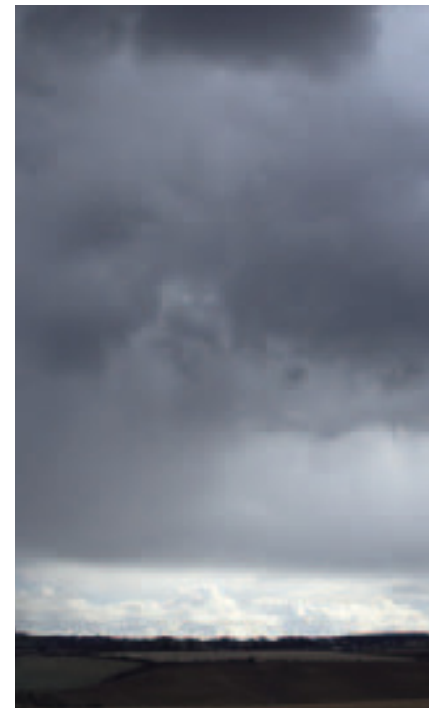
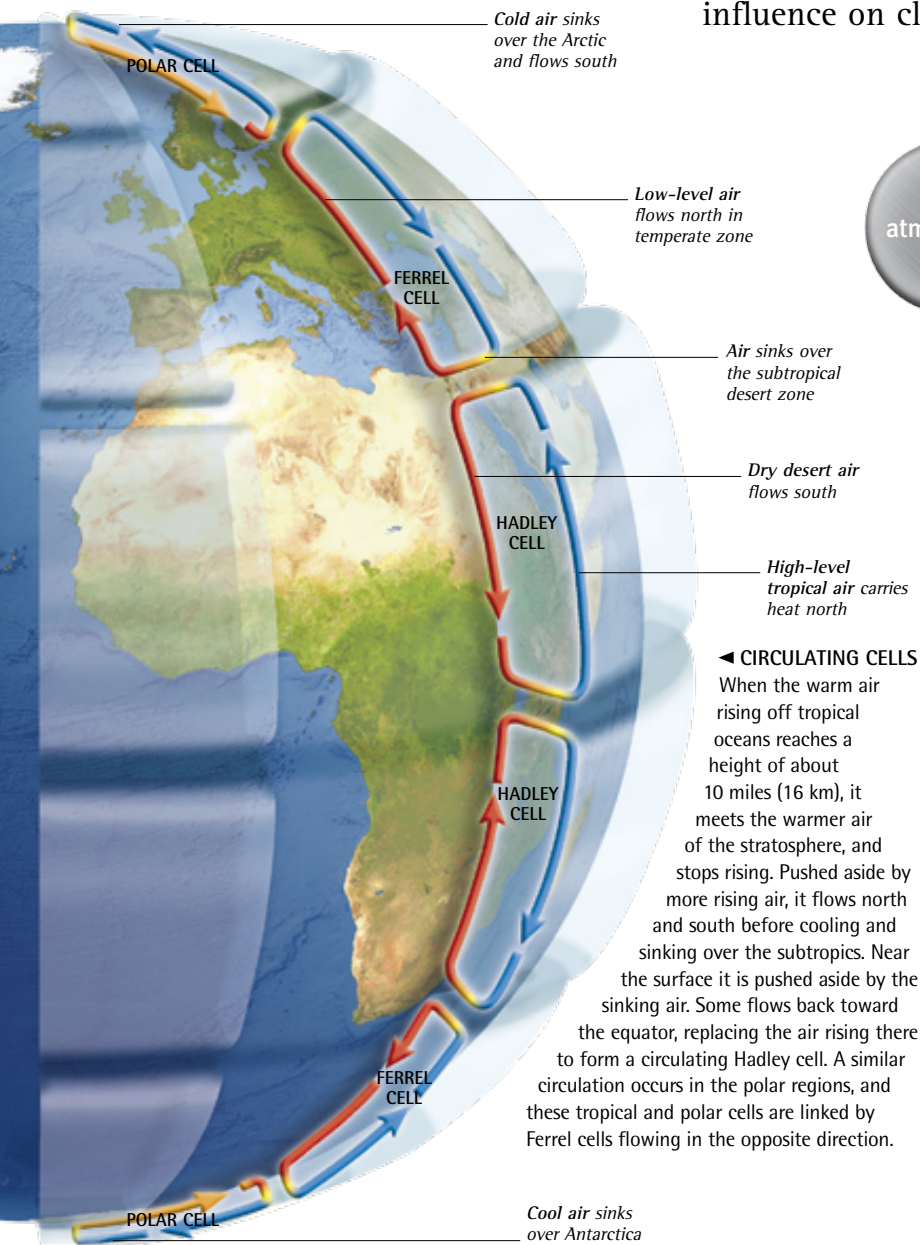


OCEANS AND THE ATMOSPHERE

The heat of the Sun makes water evaporate from oceans and rise into the air as water vapor. The vapor then cools to form clouds and rain. This evaporation effect is most intense near the equator, where it creates a zone of huge storm clouds and heavy rain. The rising air is replaced by air drawn in from farther north and south, where cooler air is sinking. This airflow creates prevailing winds that blow across the tropical oceans. The Earth's rotation makes these tropical winds swerve toward the west. Meanwhile, air drawn toward the poles from the temperate regions sweeps eastward. These prevailing winds carry weather systems with them, and in

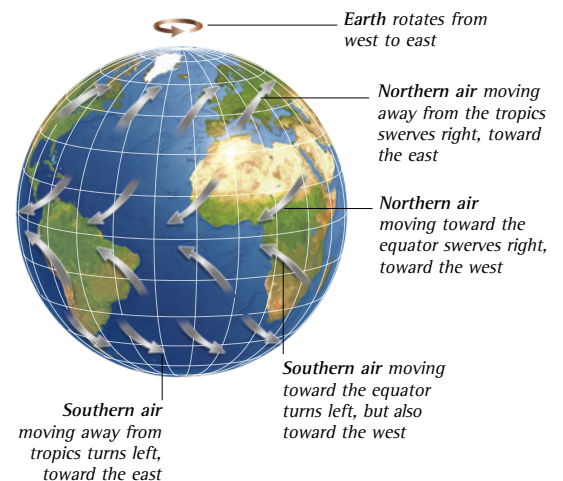
many regions they have a big influence on climate.



▲ RISING VAPOR

About 100,000 cubic miles (425,000 cubic km) of water evaporates from the oceans each year. The water vapor mixes with warm air above the oceans, which tends to rise above colder, denser air. When the air rises it expands and cools down. This makes the vapor condense into tiny water droplets that form clouds and rain.

THE SPINNING PLANET



The airflow in the lower parts of the circulating cells causes the winds. If the planet was not spinning, winds would blow directly north or south, away from the zones of sinking air toward regions where warm air is rising. But the Earth's spin pushes the airflow off-course. Air moving toward the equator swerves west, while air moving toward the poles swerves east. This is called the Coriolis effect. It creates the global pattern of prevailing winds that blow over the oceans.